

Transient Response of First Order DC Circuits	
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Roll Number: 23EE019	Date: 22-10-2024

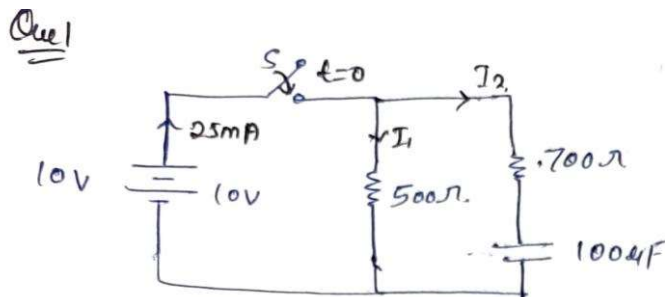
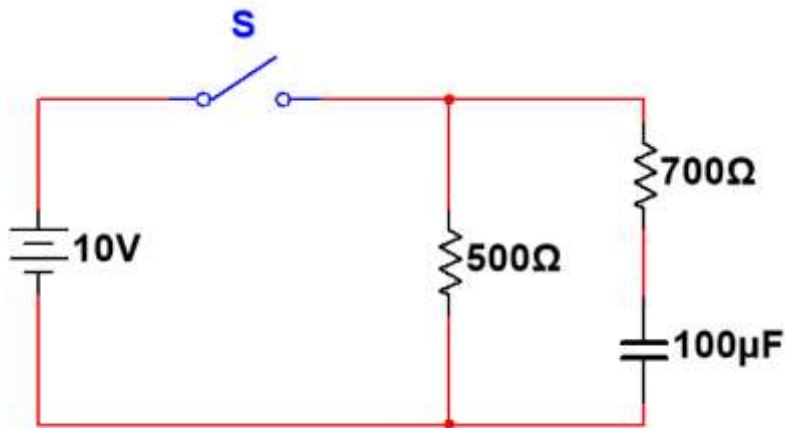
Aim:

Verify the Transient Response of First Order DC Circuits

Theoretical Calculation:

Question 1

For the circuit shown in the figure below, the switch 'S' is closed at $t=0$ sec. Determine how long will it take, after the switch is closed, for the total current draw from the supply to reach 25mA.



$$I_1(t=0) = \frac{10}{500} \text{ A}$$

$$I_2(t=0) = \frac{10}{700} \text{ A}$$

$$I_T(t=0) = \frac{10}{500} + \frac{10}{700} = 34 \text{ mA}$$

$$\boxed{I_T = 34 \text{ mA}}$$

$$t = 0^-$$

$$V(0^-) = 0V = V(0^+)$$

$$I_C(t) = \frac{V}{R} e^{-t/RC}$$

$$I_C(t=0) = \frac{10}{700} e^{-\frac{t}{200 \times 100 \times 10^{-6}}}$$

$$I_T(t) = \frac{10}{500} + \frac{10}{700} e^{-\frac{t}{7 \times 10^{-2}}}$$

$$25 \times 10^{-3} = \frac{10}{500} = \frac{10}{700} e^{-\frac{t}{7 \times 10^{-2}}}$$

$$= \frac{7}{20} = e^{-\frac{t}{7 \times 10^{-2}}}$$

$$= \ln\left(\frac{7}{20}\right) = -\frac{t}{7 \times 10^{-2}}$$

$$= -\ln\left(\frac{7}{20}\right) \times 7 \times 10^{-2} = t$$

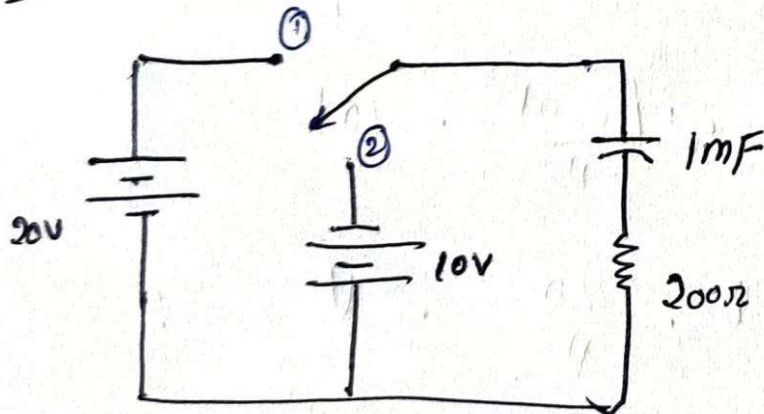
$$= 0.07348 \text{ A}$$

$$\boxed{\text{Time} = 73.5 \text{ ms.}}$$

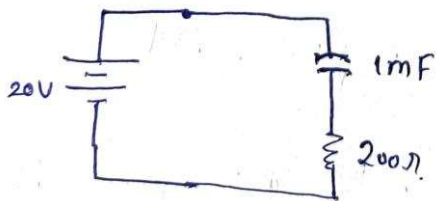
Question 2.

- In the network shown below, the switch is closed to position 1 at $t = 0$ & is moved to position 2 at $t = 0.4$ sec. Determine the voltage across the capacitor $v_C(t)$ & sketch it for $0 \leq t \leq 1$ sec
- Also find the value of 't' for which $v_C(t) = 0$

Ques 2



when connect to ①



$$\tau = RC = 200 \times 10^{-3} = 0.2 \text{ sec}$$

$$\boxed{\tau = 0.2 \text{ sec}}$$

$$V_C(t) = V(1 - e^{-t/\tau})$$

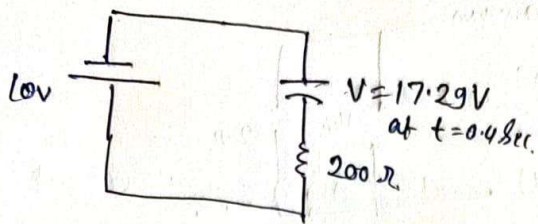
$$V_C(t) = 20(1 - e^{-t/200 \times 1 \times 10^{-3}})$$

$$V_C(t=0) = 0 \text{ V}$$

$$V_C(t=0.4) = 20(1 - e^{-0.4/0.2})$$

$$\boxed{V_C(t=0.4) = 17.29 \text{ V}}$$

When connected at ②



$$V_c(t) = V(1 - e^{-t/\tau})$$

$$V_c(t - 0.4) = -10 + 17.29(1 - e^{-\frac{(t-0.4)}{0.2}})$$

$$V_c(t - 0.4) = -$$

$$V_c(t_0) = 17.29V, \quad V(\infty) = -10V$$

$$\tau = 0.2$$

$$V(t - 0.4) = -10 - (-10 - 17.29)e^{-\frac{(t-0.4)}{0.2}}$$

$$= -10 + 27.29e^{-\frac{(t-0.4)}{0.2}}$$

$$\text{At } (t = 1.8 \text{ sec}) = -10 + 27.29e^{-\frac{(1-0.4)}{0.2}} = -8.64$$

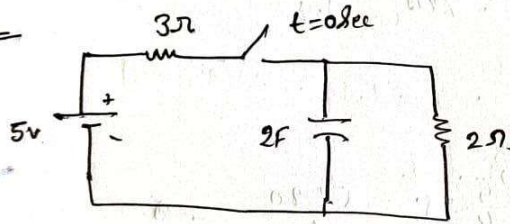
$$\text{At } t = 1.8 \text{ sec}$$

$$V_c(t) = -8.64V$$

Question 3.

- 1) Calculate initial Voltage and steady state voltage across the capacitor.
- 2) Plot voltage across capacitor and current through capacitor for $t > 0$ sec.

Que 3



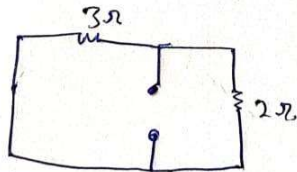
$$V(t=0) = 0V$$

$$V_{2\Omega} = \frac{2 \times 5}{5} = \frac{10}{5} = 2V$$

$$V(\infty) = 2V$$

$$V_C(t) = V_C(\infty) - [V_C(\infty) - V_C(0)] e^{-\frac{t}{R_{eq}C_{eq}}}$$

Req :



$$= \frac{3 \times 2}{3+2} = \frac{6}{5} \Omega \quad \boxed{R_{eq} = \frac{6}{5} \Omega}$$

$$V_C(0) = 0V, \quad R_{eq} = \frac{6}{5} \Omega$$

$$V_C(\infty) = 2V, \quad C_{eq} = 2F$$

$$\tau = \frac{6}{5} \times 2 = 2.4$$

$$\boxed{V_C(t=4\tau) = 1.96V}$$

$$V_C(t) = 2 - (2-0)e^{-t/\frac{6}{5} \times 2}$$

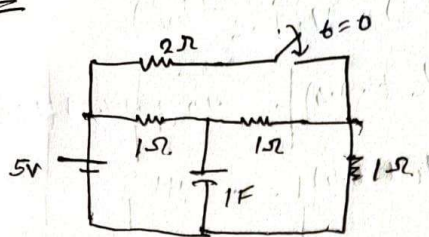
$$V_C(t) = 2 - 2e^{-5/12(t)}$$

$$V_C(t) = 2(1 - e^{-t/2.4})$$

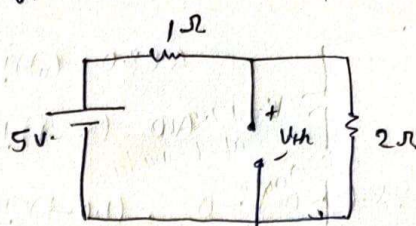
Question 4.

Que 4. In the circuit shown, the switch is closed at $t=0$ sec so calculate voltage across the capacitor for $t>0$.

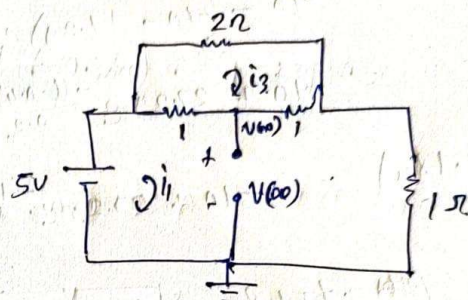
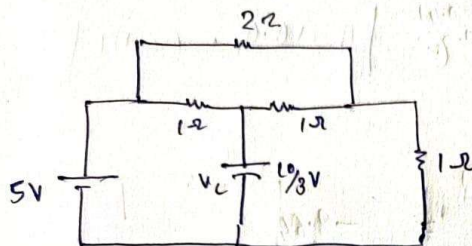
Prob 4



At $t=0^-$



At $t=0^+$

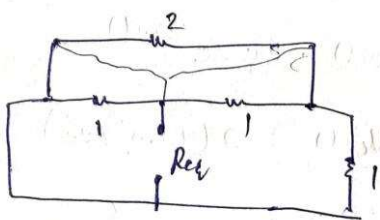


$$\begin{cases} 3i_1 - 2i_3 = 5 & \text{--- (1)} \\ -2i_1 + 4i_3 = 0 & \text{--- (2)} \end{cases} \Rightarrow \begin{cases} i_1 = 5/2 \text{ A} \\ i_3 = 5/4 \text{ A} \end{cases}$$

$$5 - v(\infty) = \left(\frac{5}{2} - \frac{5}{4}\right) \cdot 1$$

$$v(\infty) = 5 - \frac{5}{4} = \frac{20-5}{4} = \frac{15}{4} = 3.75 \text{ V}$$

Req:



$$\begin{aligned} R_{eq} &= 0.625 \Omega \\ v(\infty) &= 3.75 \end{aligned}$$

$$v(0) = \frac{10}{3} \text{ V}$$

$$V(t) = 3.75 - (3.25 - 0.33)e^{-t/0.625}$$

$$V(t) = 3.75 - (0.42)e^{-t/0.625}$$

$$\tau = 0.625$$

$$4\tau = 0.625 \times 4 = 2.5$$

$$V(t=4\tau) =$$

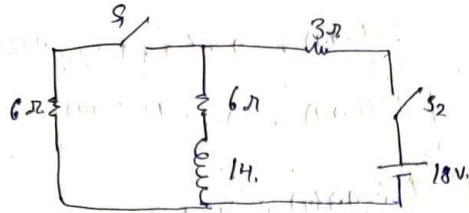
$$V(t=2.5) = 3.75 - 0.42e^{-2.5/0.625}$$

$$\boxed{V(t=2.5) = 3.74 \text{ V}}$$

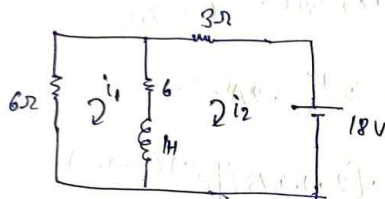
Question 5.

In the circuit shown, at $t=0$ sec S1 is closed, S2 is closed, at $t=0.1$ sec S1 is closed, S2 is open and At $T=0.3$ sec S1 is open, S2 is closed Find current through the inductor at $t=0.5$ sec.

Ques 5



$t=0$ $S_1 \rightarrow$ Closed, $S_2 \rightarrow$ closed

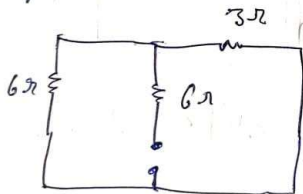


$$I_L(t=0) = 0A$$

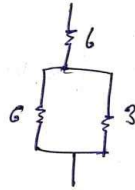
$$I_L(t=\infty) = 1.5A$$

$$\begin{cases} 12I_1 - 6I_2 = 0 \\ 9I_2 - 6I_1 = -18 \end{cases} \Rightarrow \begin{cases} I_1 = -3/2 \\ I_2 = -3 \end{cases}$$

Req = :-



Req



$$\frac{6 \times 3}{9} = \frac{18}{9} = 2\Omega$$

$$2 + 6 = 8\Omega$$

$$Req = 8\Omega$$

$$I(t) = 1.5 - (1.5 - 0)e^{-8/4t}$$

$$I(t) = 1.5 - 1.5e^{-8/4t}$$

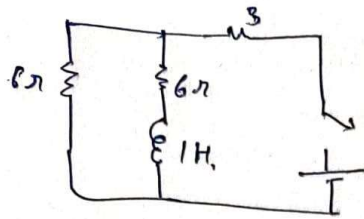
$$I(t) = 1.5(1 - e^{-8/4t})$$

$$\boxed{I(t) = 1.5(1 - e^{-8t})}$$

$$I(t=0.1) = 1.5(1 - e^{-8 \times 0.1})$$

$$I(t=0.1) = 0.83A$$

$t = 0.1 \text{ sec}$ $S_1 = \text{closed}, S_2 = \text{open}$



$$I(t_0) = 0.83 \text{ A}$$

$$I(t = \infty) = 0 \text{ A}$$

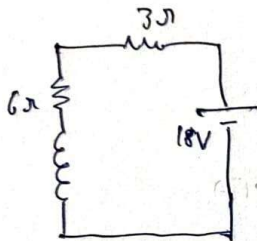
$$I(t - 0.1) = 0 - (0 - 0.83)e^{-12(t-0.1)}$$

$$I(t = 0.3) = 0.83 e^{12(0.3-0.1)}$$

$$I(t = 0.3) = 0.0753 \text{ A}$$

$$I(t - 0.1) = 0.83 e^{12(t-0.1)}$$

③ $I(t = 0.3 \text{ sec})$ $S_1 \rightarrow \text{open}, S_2 \rightarrow \text{closed}$



$$I_1(0.3) = 0.0753 \text{ A} = I(t = t_0)$$

$$I_L(\infty) = \frac{V}{R} = \frac{18}{9} = 2 \text{ A}$$

$$I(t - 0.3) = 2 - [2 - 0.0753]e^{-9(t-0.3)}$$

$$I(t - 0.3) = 2 - 1.925 e^{-9(0.5-0.3)}$$

$$I(t = 0.5) = 1.682 \text{ A}$$

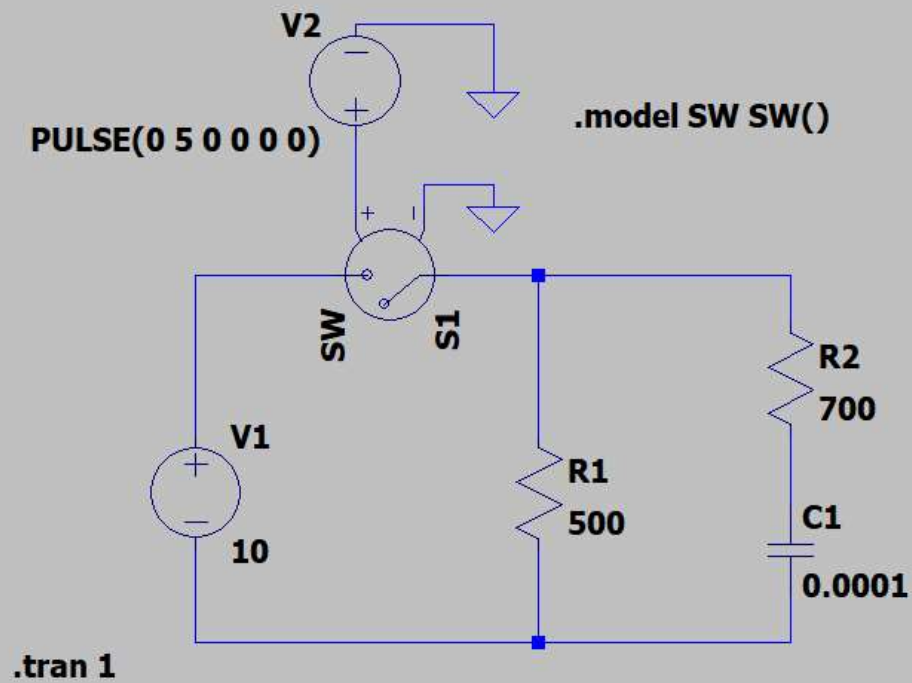
Verification by Simulation:

Question 1

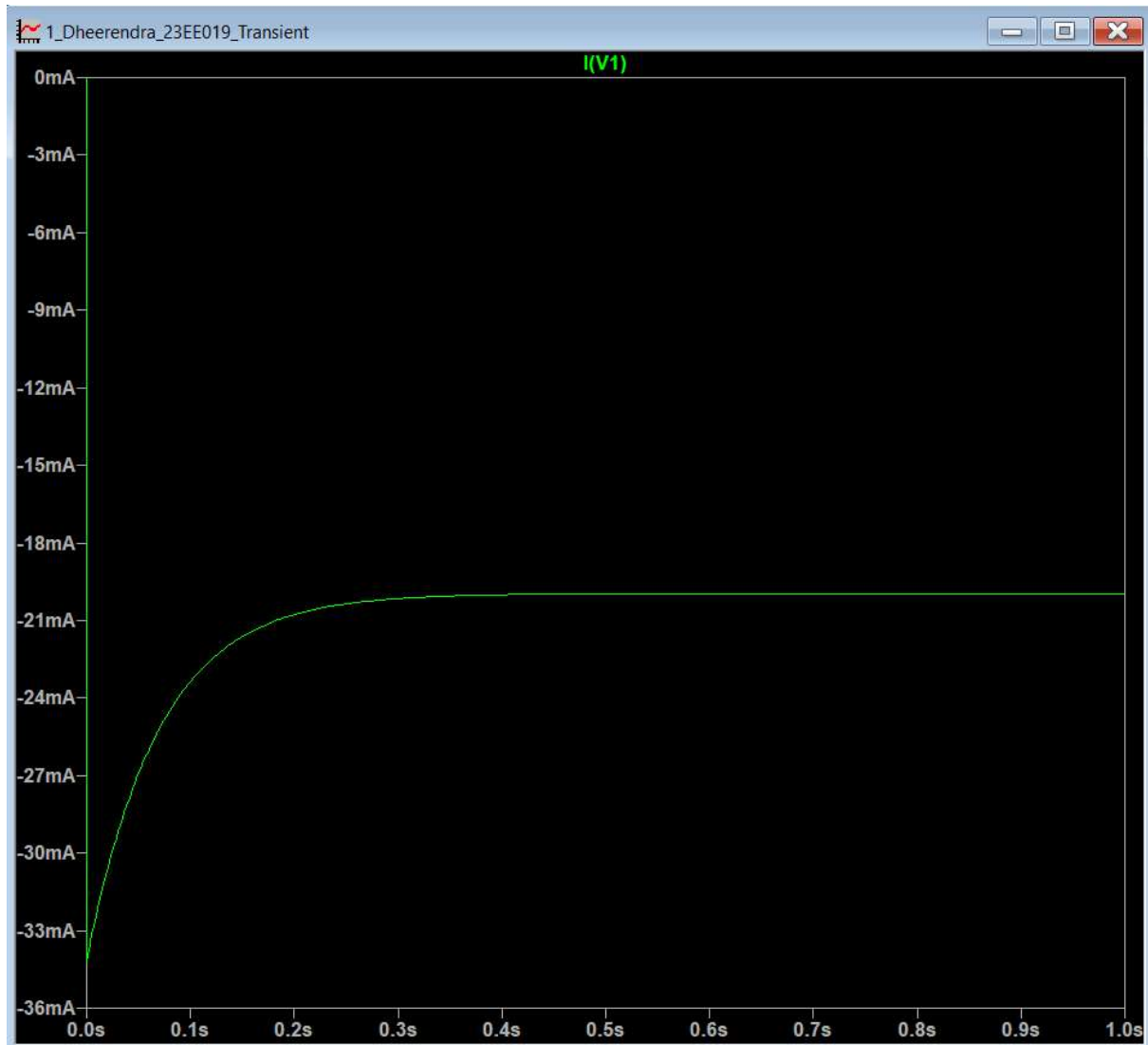
For the circuit shown in the figure below, the switch 'S' is closed at $t=0$ sec. Determine how long will it take, after the switch is closed, for the total current draw from the supply to reach 25mA.

Circuit Diagram:

Que 1. For the circuit shown in the figure below, the switch 'S' is closed at $t=0$ sec. Determine how long will it take, after the switch is closed, for the total current draw from the supply to reach 25mA.



Plot of Voltage/Current:

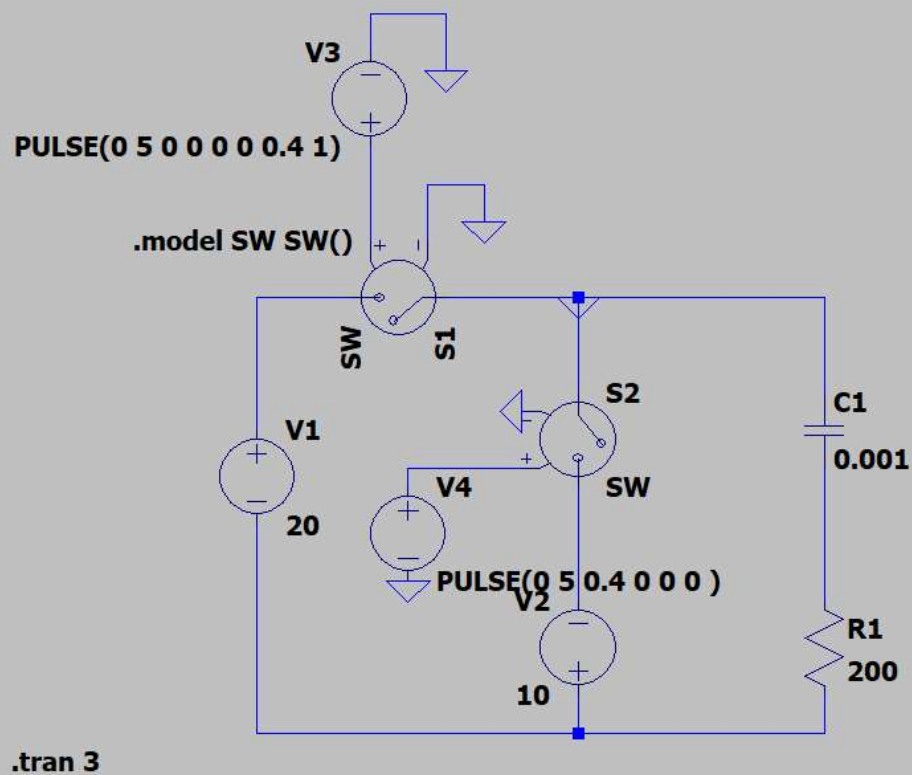


Question 2.

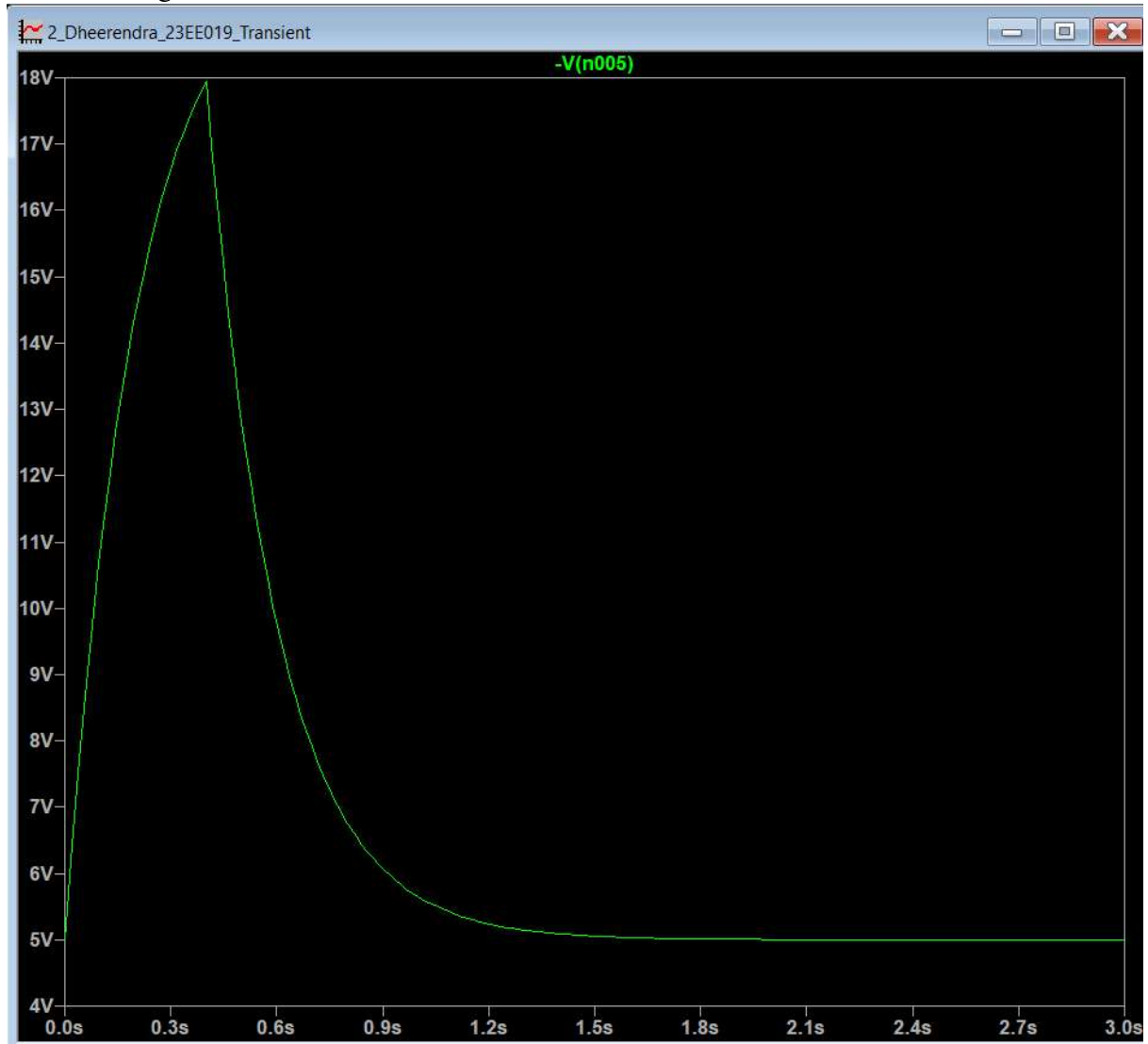
- In the network shown below, the switch is closed to position 1 at $t = 0$ & is moved to position 2 at $t = 0.4$ sec. Determine the voltage across the capacitor $v_c(t)$ & sketch it for $0 \leq t \leq 1$ sec
- Also find the value of 't' for which $v_c(t) = 0$

Circuit Diagram:

Que 2. In the network shown below, the switch is closed to o position 1 at $t=0$ & is moved to position 2 at $t=0.4$ sec. Determine the voltage across the capacitor $V_c(t)$ & sketch it for $0 < t < 1$ sec Also find the value of 't' for which $(t)=0$



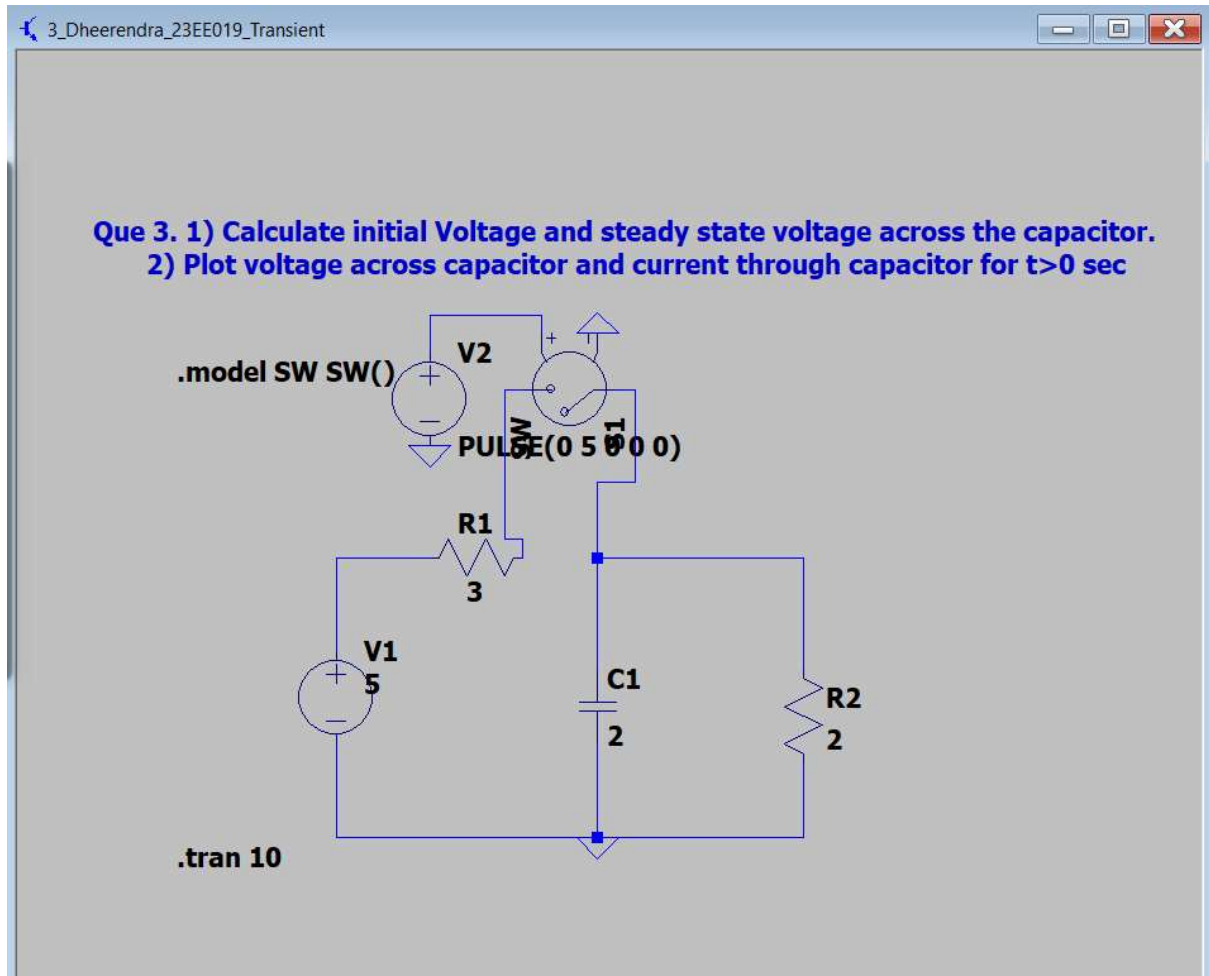
Plot of Voltage/Current:



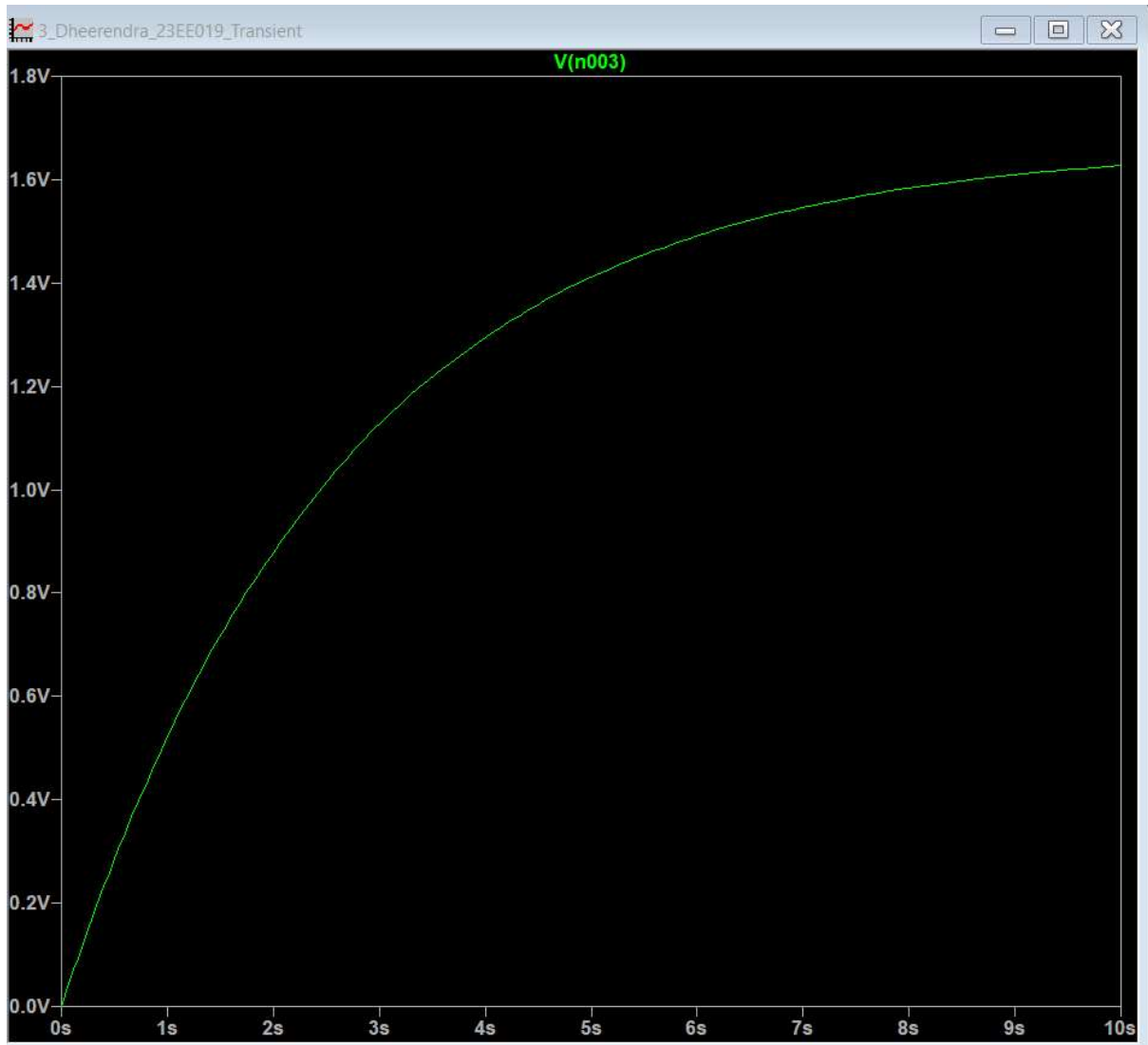
Question 3.

- 1) Calculate initial Voltage and steady state voltage across the capacitor.
- 2) Plot voltage across capacitor and current through capacitor for $t \geq 0$ sec.

Circuit Diagram:



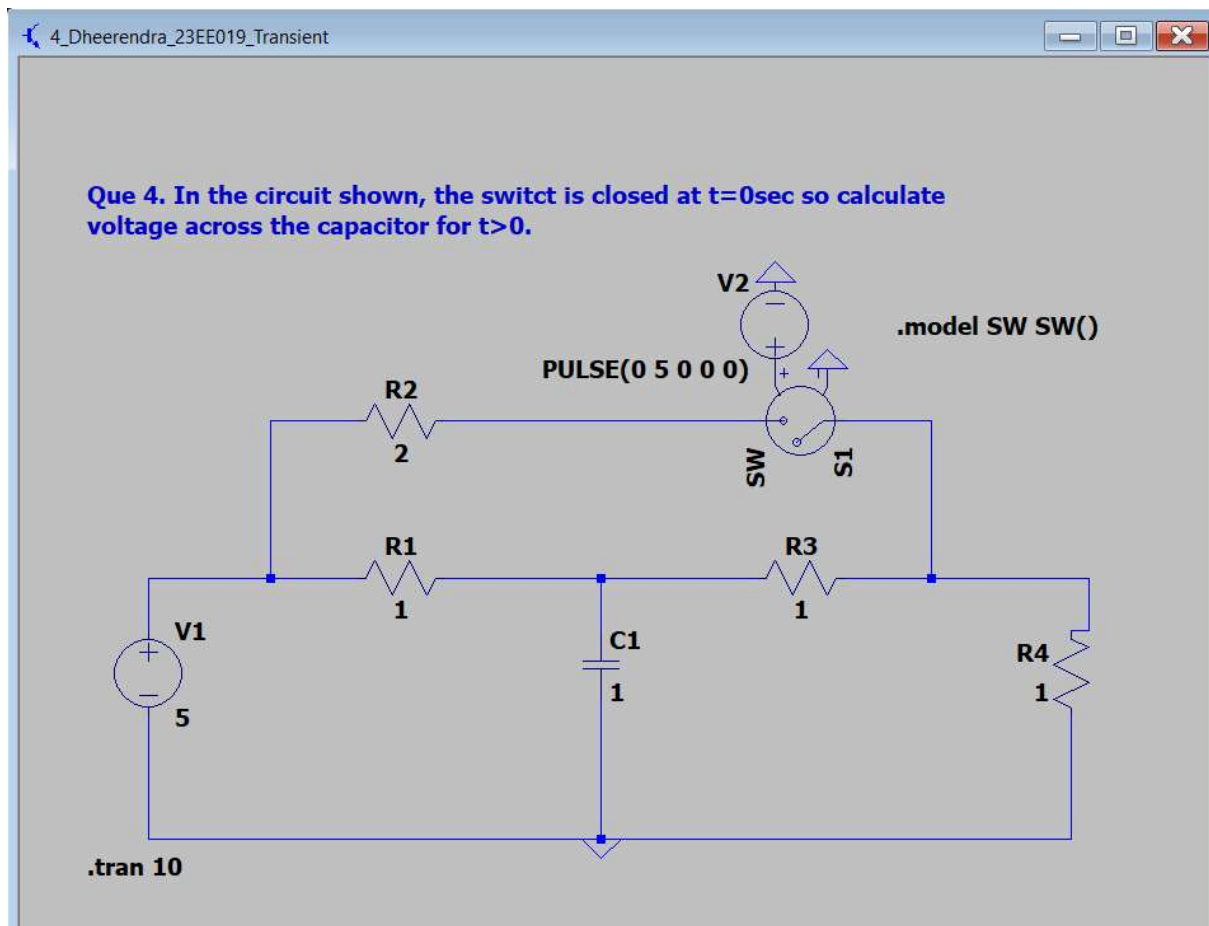
Plot of Voltage/Current:



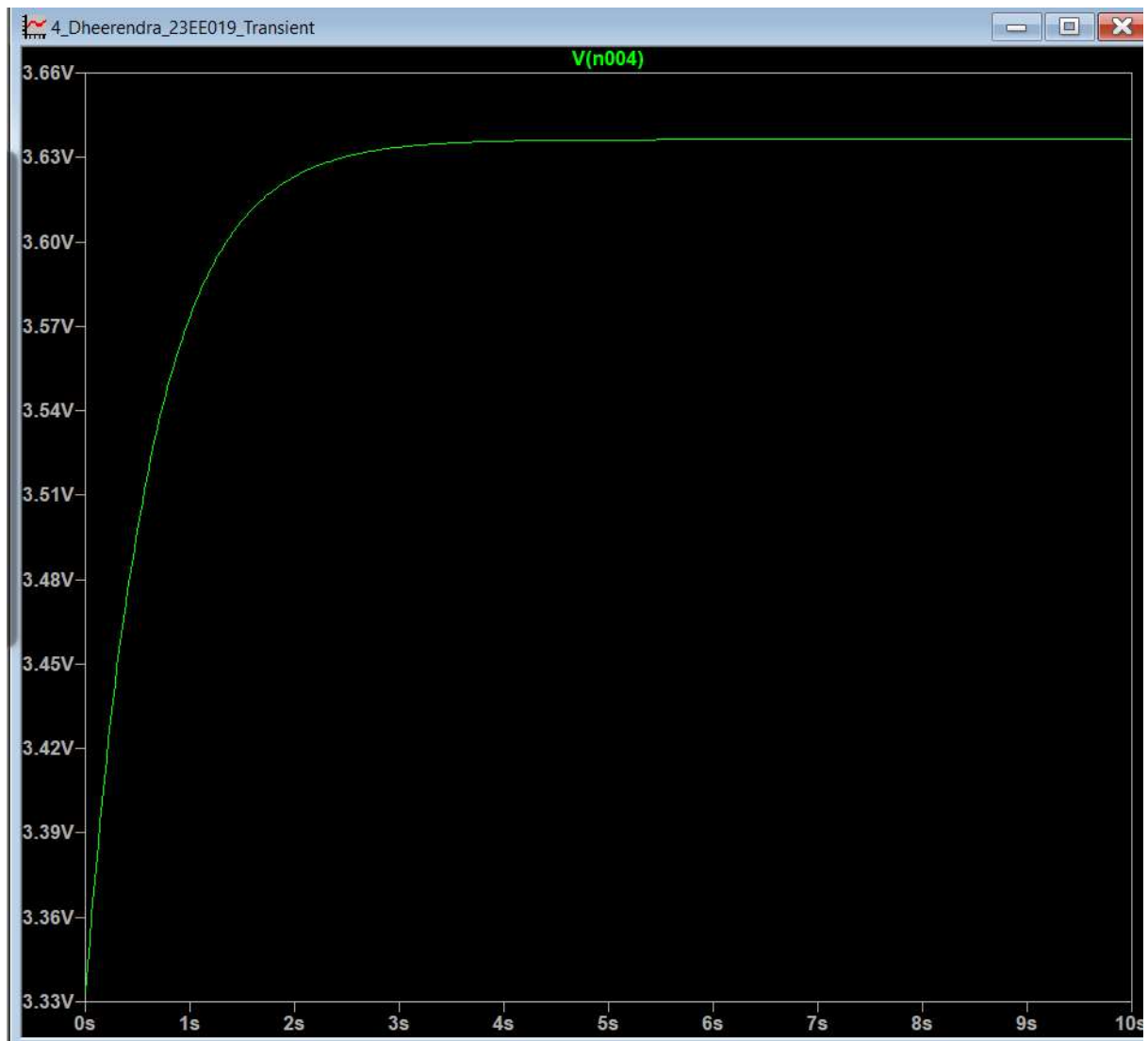
Question 4.

Que 4. In the circuit shown, the switch is closed at $t=0\text{sec}$ so calculate voltage across the capacitor for $t>0$.

Circuit Diagram:



Plot of Voltage/Current:

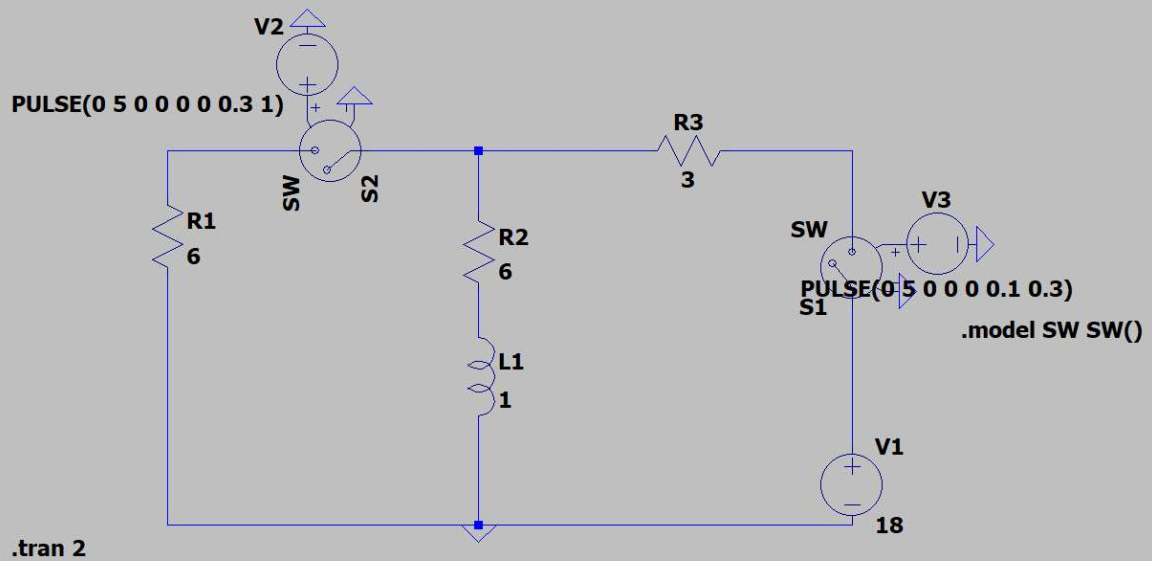


Question 5.

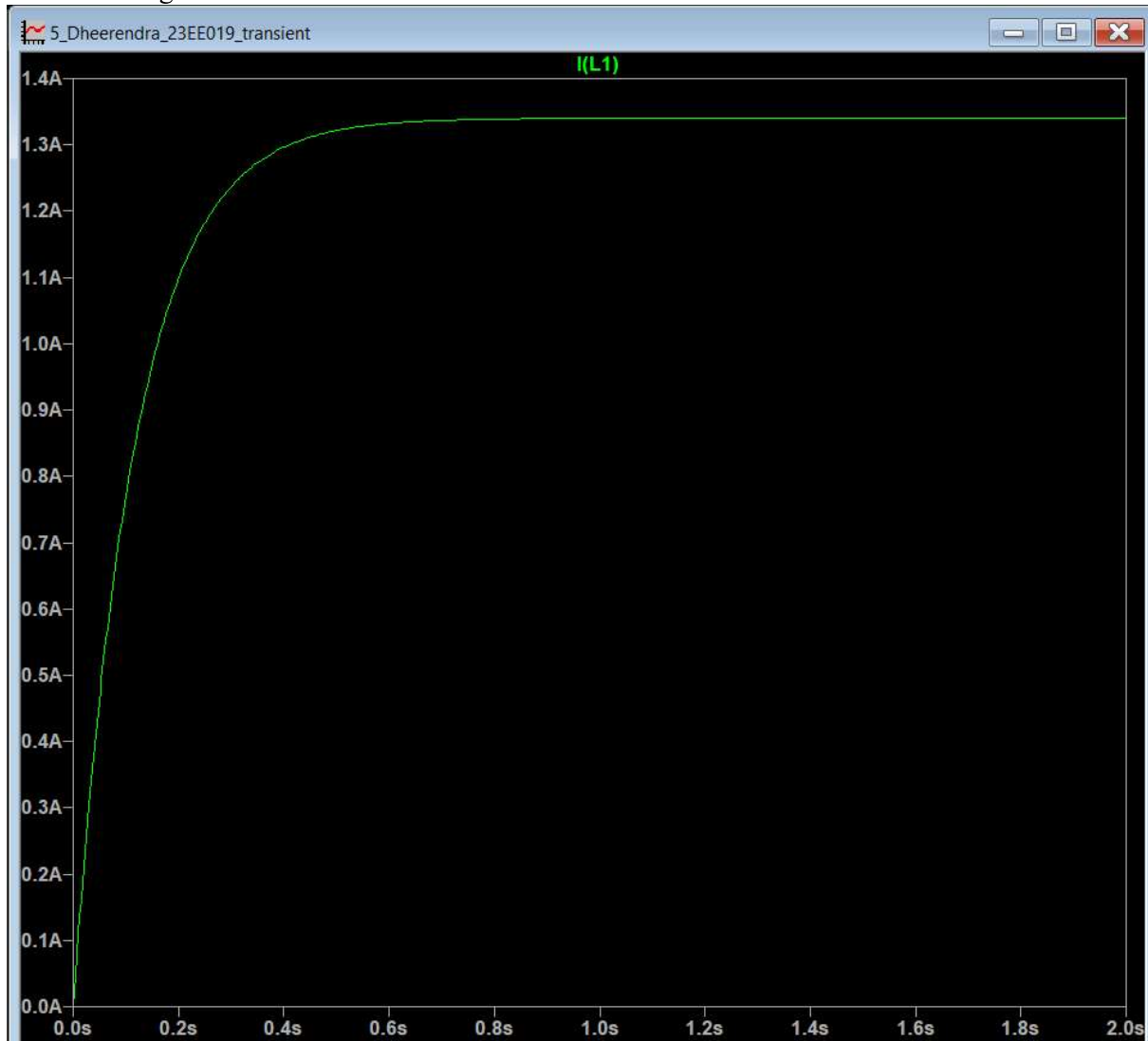
In the circuit shown, at $t=0$ sec S1 is closed, S2 is closed, at $t=0.1$ sec S1 is closed, S2 is open and At $T=0.3$ sec S1 is open, S2 is closed Find current through the inductor at $t=0.5$ sec.

Circuit Diagram:

Que 5. In the circuit shown, At $t=0$ sec S1 is closed, S2 is closed,
 at $t=0.1$ sec S1 is closed, S2 is open and
 At $T=0.3$ sec S1 is open , S2 is closed
 Find current through the inductor at $t=0.5$ sec



Plot of Voltage/Current:



Observation Table:

S. No.	$V_c(t)$	$I_L(t)$	Practical Value	Theoretical Value
1	-	-	Time = 73.438214ms	Time = 73.5ms
2.(i)	$V_c(t) = 20(1 - e^{-(t/0.2)})$	-	$V_c(0.4) = 17.65V$	$V_c(0.4) = 17.29V$
2.(ii)	$V_c(t-0.4) = -10 + 27.29(e^{-(t-0.4)/0.2})$	-	$V_c(1) = -8.74V$	$V_c(1) = -8.64V$
3	$V_c(t) = 2(1 - e^{-(5t/12)})$	-	$V_c(t=4\tau) = 1.6238882V$	$V_c(t=4\tau) = 1.96V$

4	$V_c(t) = 3.75 - 0.42e^{(-t/0.625)}$	-	$V_c(t=4\tau) = 3.630V$	$V_c(t=4\tau) = 3.74V$
5. (i)	-	$I(t) = 1.5 (1 - e^{-8t})$	$I(0.1) = 0.772A$	$I(0.1) = 0.83A$
5. (ii)	-	$I(t-0.1) = 0.83(e^{-12(t-0.1)})$	$I(0.3) = 0.0653A$	$I(0.3) = 0.0753A$
5. (iii)	-	$I(t-0.3) = 2 - 0.0753e^{(-9(t-0.3))}$	$I(0.5) = 1.582A$	$I(0.5) = 1.682A$

Results and Conclusion:

As we see that the theoretical Values is approximately equal to the practical values, Hence Transient Response of First Order DC Circuits is Verified.